

The use of the ghost fluid method for Poisson's equation to simulate streamer propagation in point-to-plane and point-to-point geometries

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Received 6 October 2008, in final form 22 December 2008

Published 26 February 2009

Online at stacks.iop.org/JPhysD/42/065203

Abstract

This paper presents the application of the ghost fluid method (GFM) to solve Poisson's equation for streamer discharge simulations between electrodes of complex geometries. This approach allows one to use a simple rectilinear grid and nevertheless take into account the influence of the exact shape of the electrode on the calculation of the potential and the electric field. First, the validity of the GFM approach concerning the computation of the electric field is demonstrated by performing direct comparisons in a point-to-plane geometry of the Laplacian potential and electric field calculated with this method and given by the analytical solution. Second, the GFM is applied to the simulation of a positive streamer propagation in a hyperboloid-to-plane configuration studied by Kulikovsky (1998 *Phys. Rev. E* **57** 7066–74). Very good agreement is obtained with the results of Kulikovsky (1998) on all positive streamer characteristics during its propagation in the interelectrode gap. Then the GFM is applied to simulate the discharge in preheated air at atmospheric pressure in point-to-point geometry. The propagation of positive and negative streamers from both point electrodes is observed. After the interaction of both discharges, the very rapid propagation of the positive streamer towards the cathode in the volume pre-ionized by the negative streamer is presented. This structure of the discharge is in qualitative agreement with the experiment.

(Some figures in this article are in colour only in the electronic version)

1. Introduction

Low temperature plasma discharges operating at atmospheric pressure have been receiving increasing attention in recent years (Roth 1995, Kunhardt 2000, Kogelschatz *et al* 1999,

Fridman *et al* 2005). They are studied for an increasing list of applications such as ozone generation, pollution control, biological decontamination, plasma assisted combustion, air flow control and thin film coatings. In air at atmospheric pressure, discharges usually take the form of thin plasma filaments, also called streamers. In parallel to experimental studies, numerous simulation works have been done to better understand the properties of streamers. Many early

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simulation studies have been done in the simple plane-to-plane geometry (Dhali and Williams 1987, Vitello *et al* 1994, Kulikovsky 1997b). However, in practical situations, the electrode geometries are much more complex with often one of the electrodes having a radius of curvature much smaller than the other (e.g. point-to-plane, wire-cylinder). For streamer simulations, one of the most studied configurations in the literature is the point-to-plane geometry. For the sphere-plane configuration, Babaeva and Naidis (1996) have proposed an approach to take into account the influence of the spherical electrode on the discharge without including it in the computational domain. In this model, the computational grid is tangent to the sphere and therefore there is only one point of contact between the spherical electrode and the computational domain. The simulation of the discharge close to the sphere, in regions where the computational grid has no contact with the electrode, is roughly approximated. It is interesting to note that this approach is rather easy to implement and has been used in different works in recent years (e.g. Liu and Pasko (2006), Liu *et al* (2007)). The only complexity of this model is the implementation of boundary conditions for Poisson's equation where image charges in the sphere have to be taken into account. Recently a similar approach has been used by Luque *et al* (2008) to study streamer propagation in a point-to-plane geometry without including the point electrode in the computational domain. In this approach the needle is approximated by a cylinder of given length and the tip of the electrode is assumed to have a constant radius. In this approach, as in the work of Babaeva and Naidis (1996) there is only one point of contact between the point electrode and the computational domain.

For more realistic electrode shapes, as for hyperboloids (Kulikovsky 1997b, Morrow and Lowke 1997, Pancheshnyi *et al* 2001) or paraboloids (Djermoune *et al* 1995a, 1995b, Dessantes 2000), the electrode is included in the computational domain. In these works, streamer simulations are carried out using a finite volume approach and a rectilinear grid. Then, in the vicinity of the point electrode, even with a refined grid, the shape of the electrode is approximated by a staircase. The influence of this approximation on the streamer simulations is not discussed in the literature. It is interesting to note that in the works of Morrow and Lowke (1997), Djermoune *et al* (1995a, 1995b) and Dessantes (2000), the shape of the electrode is included as accurately as possible by defining a new axial grid point for each radial grid point such that the position defined by the axial and radial coordinates lies on the electrode surface. This definition of the grid allows one to define very accurately the potential on the electrode surface and then the boundary condition for Poisson's equation. However, the use of this approach for hyperboloids and paraboloids leads to significant variations of the cell sizes to adapt the point electrode shape. In finite volume methods, these significant variations of cell sizes are not recommended. Furthermore, with this approach, a new grid has to be generated for any change in the electrode geometry. This is a major limitation to easily carry out parametric studies on the influence of the electrode geometry on the discharge structure.

Another approach is to use a curvilinear grid (Kim *et al* 2004, Akishev *et al* 2002). For example, for an hyperboloid-to-plane geometry, a quasi-orthogonal grid can be defined based on confocal hyperboloids and ellipses. In this case, the grid is orthogonal to the electrode surface and the potential of the electrode is set very accurately. Furthermore, with this approach, it is rather easy to implement boundary conditions for charged species with various secondary emission processes. However, the difficulty with curvilinear grids is the stretching of the grid spacing which can reduce the resolution, unless multiple grids with multi-resolution patching techniques are used.

Finally, some authors (Georghiou 1999, Papageorghiou and Spyrou 2007) use finite element methods with unstructured grids to simulate streamer propagation in wire-plate or point-to-plane geometries. It is clear that the use of an unstructured grid allows an accurate local description of complex geometries with a reduced number of points in comparison with structured grids. Recently, Ducasse *et al* (2007) have compared the results obtained with finite volume and finite element approaches with, respectively, structured and unstructured grids for a point-to-plane geometry with a 1 mm interelectrode gap and a hyperboloid anode. A good agreement between the results obtained with both methods has been observed. However, it is important to note that the implementation of the finite element method is much more complex than the finite volume approach. Ducasse *et al* (2007) have also shown that even with a smaller number of nodes in the grid, the computational cost of finite element method calculations appeared to be much higher than the one of finite volume method calculations.

In this work, we have implemented a method allowing us to easily take into account electrodes of complex geometries in streamer simulations. For simplicity, we have tried to find a method based on rectilinear grids and finite volume or difference approaches. The problem of taking into account a boundary or an interface of complex geometry in a rectilinear grid has been studied in fluid mechanics. For example, to model interfaces in two-phase flows, Fedkiw *et al* (1999) have introduced the ghost fluid method (GFM). This method allows one to localize accurately and to follow the time evolution of the interface with a level set formalism (Liu *et al* 2000) and to accurately define boundary conditions at the interface. Liu *et al* (2000) developed the method for the variable coefficient Poisson's equation. Kang *et al* (2000) extended it to treat two-phase incompressible flow including the effects of viscosity, surface tension and gravity. Nguyen *et al* (2001) included the reactivity of the flows. For Dirichlet boundary conditions set on the interface the second-order-accurate symmetric discretization of Poisson's equation based on the GFM was obtained in Gibou *et al* (2002). A review of boundary condition capturing methods based on the GFM is provided in Hong *et al* (2007).

Preliminary tests to apply the GFM to the simulation of a streamer discharge have been done by Régnier (2004). In contrast to the two-phase flows, the situation is simpler for streamer simulations since the embedded boundary (the electrode) is motionless. In this work, we adapted the GFM from Gibou *et al* (2002) to the simulation of streamer

discharge propagation in classical point-to-plane and point-to-point geometries.

In section 2, we present the equations needed for the numerical simulation of streamer discharges and the GFM applied to Poisson's equation. In section 3, to validate the use of the GFM for the calculation of the Laplacian electric potential and the electric field, we compare the results obtained using this approach with the analytical solution in a point-to-plane configuration. Then, we apply the GFM to the simulation of a positive streamer propagation in a hyperboloid-to-plane configuration studied by Kulikovsky (1998) and we compare both results. Finally, the application of the GFM to the simulation of a discharge in preheated air in point-to-point geometry is presented and discussed.

2. Model formulation

The most common and effective model to study the dynamics of streamers is based on the following drift–diffusion equations for electrons and ions coupled with Poisson's equation (e.g. Kulikovsky 1997b):

$$\frac{\partial n_e}{\partial t} + \vec{\nabla} \cdot n_e \vec{v}_e - \vec{\nabla} \cdot (\overline{\overline{D_e}} \vec{\nabla} n_e) = S_{ph} + S_e^+ - S_e^-, \quad (1)$$

$$\frac{\partial n_p}{\partial t} = S_{ph} + S_p^+ - S_p^-, \quad (2)$$

$$\frac{\partial n_n}{\partial t} = S_n^+ - S_n^-, \quad (3)$$

$$\nabla^2 V = -\frac{q_e}{\epsilon_0} (n_p + n_n - n_e) = -\frac{\rho}{\epsilon_0}, \quad (4)$$

where subscripts 'e', 'p' and 'n' refer to electrons, positive and negative ions, respectively, n_i is the number density of species i , q_e is the absolute value of the elementary charge, ρ is the net charge density, V is the electric potential, $\vec{v}_e = -\mu_e \vec{E}$ ($\vec{E}$ being the electric field) is the drift velocity of electrons, $\overline{\overline{D_e}}$ and μ_e are the diffusion tensor and the absolute value of mobility of electrons, respectively, and ϵ_0 is the permittivity of free space. On timescales of interest for the studies presented in this paper, ions are assumed to be motionless. The S^+ and S^- terms stand for the rates of production and loss of charged particles. The S_{ph} term is the rate of electron–ion pair production due to photoionization in the gas. In this paper, all transport parameters and reaction rates in air are assumed to be functions of the local reduced electric field E/N , where E is the electric field magnitude and N is the air neutral density, and are taken from Kulikovsky (1997b). In this paper axisymmetric streamers are studied and then cylindrical coordinates are used.

It is important to note that for streamer simulations, the electric field is a key parameter since, first, transport parameters and source terms in equations (1)–(3) have a strong non-linear dependence on it and second, the electric field is

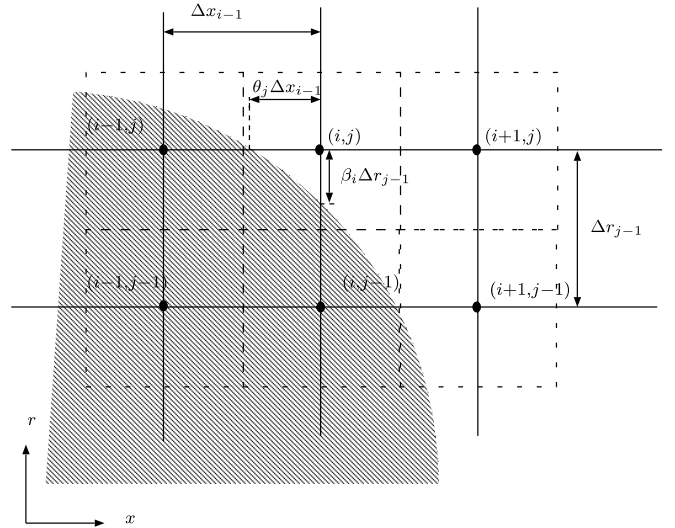


Figure 1. Electrode embedded in a Cartesian grid. The electric potential is fixed in the electrode (hatched zone) and computed in the fluid domain (non-hatched zone). Nodes are represented on the grid and are indexed by (i, j) . Dashed lines are interfaces of the grid cells.

directly related to charged species densities in equation (4). Therefore the simulation of streamer propagation requires an accurate calculation of the electric field in the simulation domain at each timestep, especially in regions with high field gradients (e.g. the streamer head and the tip of the needle electrode). In the following section, we present the application of the GFM to Poisson's equation to calculate accurately the potential and the electric field in streamer simulations.

2.1. The GFM applied to Poisson's equation

In cylindrical coordinates, equation (4) can be written as

$$\frac{\partial}{\partial x} \left(r \frac{\partial V}{\partial x} \right) + \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) = -\frac{\rho(x, r)}{\epsilon_0}, \quad (5)$$

where x and r are axial and radial coordinates, respectively. In this work, we consider that the computational domain is discretized on a rectilinear grid. The nodes of the grid are located at the centre of each control volume and are indexed with i and j in the axial and radial directions, respectively (see figure 1), such as $V_{i,j} = V(x_i, r_j)$. The edges of the cell indexed (i, j) are located by $x_{i \pm 1/2}$ and $r_{j \pm 1/2}$, in the axial and the radial directions, respectively. In the volume of the computational domain (i.e. far from the boundaries) the second-order finite difference discretization of equation (5) in cylindrical coordinates gives the classical five diagonal linear system:

$$\begin{aligned} & V_{i,j}^e V_{i+1,j} + V_{i,j}^w V_{i-1,j} + V_{i,j}^s V_{i,j-1} + V_{i,j}^n V_{i,j+1} + V_{i,j}^c V_{i,j} \\ & = -\frac{\rho_{i,j}}{\epsilon_0}, \end{aligned} \quad (6)$$

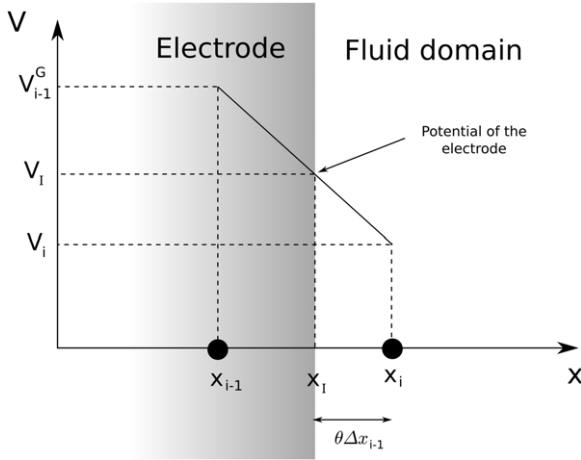


Figure 2. 1D description of the GFM applied to the calculation of the electrical potential. A virtual (or ghost) potential V_{i-1}^G is set on the node $i-1$ inside the electrode such that the interpolated potential at the interface I between the two nodes $i-1$ and i is precisely the physical potential of the electrode V_I .

where

$$\begin{cases} V_{i,j}^e = \frac{1}{\Delta x_i(x_{i+1/2} - x_{i-1/2})}, \\ V_{i,j}^w = \frac{1}{\Delta x_{i-1}(x_{i+1/2} - x_{i-1/2})}, \\ V_{i,j}^n = \frac{2r_{j+1/2}}{\Delta r_j(r_{j+1/2}^2 - r_{j-1/2}^2)}, \\ V_{i,j}^s = \frac{2r_{j-1/2}}{\Delta r_{j-1}(r_{j+1/2}^2 - r_{j-1/2}^2)}, \\ V_{i,j}^c = -(V_{i,j}^w + V_{i,j}^e + V_{i,j}^s + V_{i,j}^n), \end{cases} \quad (7)$$

with $\Delta x_i = x_{i+1} - x_i$ and $\Delta r_j = r_{j+1} - r_j$. As shown in figure 1, the electrode is an embedded boundary in the Cartesian grid. To localize accurately the fluid/electrode interface, we use two ‘signed distance functions’ based on the level set formalism (see Liu *et al* (2000)): $\phi_{i,j}^x$ is defined by the distance between the grid nodes and the electrode interface according to x , and $\phi_{i,j}^r$ is defined by the distance between the grid nodes and the electrode interface according to r . In figure 1, we define θ_j and β_i such that $|\phi_{i,j}^x| = \theta_j \Delta x_{i-1}$ and $|\phi_{i,j}^r| = \beta_i \Delta r_{j-1}$. The distance functions are readily found if one has an analytical expression of the electrode interface. Distance functions are defined as negative for nodes inside the electrode and as positive for nodes in the computational domain. For example, in figure 1, the node (i, j) is outside the electrode, and therefore the distance functions (or level set functions) are defined as positive for this node. Through the fluid/electrode interface, the potential is continuous, but there is a jump in the electric field. The GFM consists of setting virtual potentials (denoted by the superscript ‘G’ for ‘ghost’ in the following) on the nodes inside the electrode such that the interpolated potential at the interface between the two nodes of the grid is precisely the physical potential of the electrode.

To show the new discretization of Poisson’s equation around the electrode interface using the GFM, we consider as an example the case depicted by figure 1 (with an interface

lying between the nodes $(i-1, j)$ and (i, j) in the axial direction and $(i, j-1)$ and (i, j) in the radial direction). In this case, the nodes $(i-1, j)$ and $(i, j-1)$ are ‘ghost’ nodes, and then equation (6) becomes

$$V_{i,j}^e V_{i+1,j} + V_{i,j}^w V_{i-1,j}^G + V_{i,j}^s V_{i,j-1}^G + V_{i,j}^n V_{i,j+1} - (V_{i,j}^w + V_{i,j}^e + V_{i,j}^s + V_{i,j}^n) V_{i,j} = -\frac{\rho_{i,j}}{\epsilon_0}. \quad (8)$$

To derive the ghost values, as proposed by Gibou *et al* (2002), we have used a 1D extrapolation split in x and r . First, we propose to describe the method in the axial direction. Figure 2 shows the corresponding 1D configuration with an interface between nodes $i-1$ and i and with V_{i-1}^G at the $i-1$ node. In Gibou *et al* (2002) different types of extrapolation are proposed to derive the value of $V_{i-1,j}^G$. In this work, we write $V_{i-1,j}^G$ as a linear function of x_{i-1} as

$$\begin{aligned} V_{i-1,j}^G &= \frac{V_{i,j} - V_I}{|\phi_{i,j}^x|} (x_{i-1} - x_I) + V_I \\ &= -\frac{|\phi_{i-1,j}^x|}{|\phi_{i,j}^x|} (V_{i,j} - V_I) + V_I, \end{aligned} \quad (9)$$

where x_I is the position of the interface. Using equation (9) in equation (8), the terms which are factors of $V_{i,j}^w$ become

$$\begin{aligned} -V_{i,j}^w V_{i,j} + V_{i,j}^w V_{i-1,j}^G &= V_{i,j}^w (V_I - V_{i,j}) \frac{|\phi_{i,j}^x| + |\phi_{i-1,j}^x|}{|\phi_{i,j}^x|} \\ &= -\frac{V_{i,j}^w}{\theta_j} V_{i,j} + \frac{V_{i,j}^w}{\theta_j} V_I. \end{aligned} \quad (10)$$

All other terms in equation (8) remain unchanged. A similar approach can be applied in the radial direction for the interface lying between the nodes $(i, j-1)$ and (i, j) . Finally, for the 2D case of figure 1, one can easily show that the discretization of Poisson’s equation centred on the point (i, j) becomes

$$\begin{aligned} V_{i,j}^{e,g} V_{i+1,j} + V_{i,j}^{w,g} V_{i-1,j} + V_{i,j}^{s,g} V_{i,j-1} + V_{i,j}^{n,g} V_{i,j+1} \\ + V_{i,j}^{c,g} V_{i,j} = S_{i,j}^g, \end{aligned} \quad (11)$$

where

$$\begin{cases} V_{i,j}^{e,g} = V_{i,j}^e, \\ V_{i,j}^{w,g} = 0, \\ V_{i,j}^{n,g} = V_{i,j}^n, \\ V_{i,j}^{s,g} = 0, \\ V_{i,j}^{c,g} = -V_{i,j}^e - \frac{1}{\theta_j \Delta x_{i-1}(x_{i+1/2} - x_{i-1/2})} \\ \quad - V_{i,j}^n - \frac{2r_{j-1/2}}{\beta_j \Delta r_{j-1}(r_{j+1/2}^2 - r_{j-1/2}^2)}, \\ S_{i,j}^g = -\frac{\rho_{i,j}}{\epsilon_0} - \frac{V_I}{\theta_j \Delta x_{i-1}(x_{i+1/2} - x_{i-1/2})} \\ \quad - \frac{2r_{j+1/2} V_I}{\beta_j \Delta r_j(r_{j+1/2}^2 - r_{j-1/2}^2)}, \end{cases} \quad (12)$$

and where the superscript ‘g’ is used to define coefficients of Poisson’s equation close to the interface. Different situations have to be considered depending on how the electrode crosses

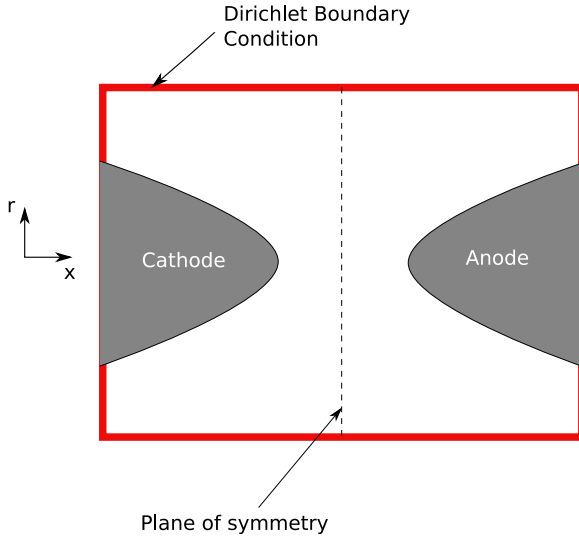


Figure 3. Point-to-point configuration. Boundary conditions at the electrode surface are taken into account using the GFM. The thick line shows the borders of the computational domain. The configuration is axisymmetric and also has a planar symmetry.

the cells. This requires to check which of the five nodes required for the discretization of Poisson's equation lie in the electrode. This can be easily done by testing the signs of either ϕ^x or ϕ^r . For example, if one finds a negative value of $\phi_{i,j}^x$ at the central node (i, j) , one directly sets $V_{i,j} = V_I$. In the case of an interface located between two nodes in the radial direction *only*, then $V_{i,j}^e$ and $V_{i,j}^w$ remain unchanged and are given by equation (7), as well as $V_{i,j}^s$, if the electrode is above the grid node (i, j) , or $V_{i,j}^n$, if the electrode is below. In the case of an interface located between two nodes in the axial direction *only*, then $V_{i,j}^n$ and $V_{i,j}^s$ remain unchanged and are given by equation (7), as well as $V_{i,j}^w$, if the electrode is to the right of the grid node (i, j) , or $V_{i,j}^e$, if the electrode is to the left.

In this work, we consider streamer simulations in point-to-plane and point-to-point geometries. The GFM is used to define the electric potential on the electrodes but boundary conditions have to be defined on the other boundaries of the computational domain. For the point-to-point geometry, figure 3 shows the two hyperboloid electrodes and the borders of the computational domain where boundary conditions have to be defined. Usually, a large computational domain is used in simulations, and the electric potential at these boundaries is specified by neglecting contributions from the charge inside the domain due to the relatively small space occupied by the charge. In this case, two different types of boundary conditions can be used. First, the homogeneous Neumann boundary condition ($\vec{\nabla} V \cdot \vec{n} = 0$, $\vec{n}$ being the normal vector of the surface boundary) and, second, the Dirichlet boundary condition based on the solution of Laplace's equation (i.e. $\rho = 0$ in equations (4) and (5)). For point-to-plane and point-to-point geometries, we have compared the results obtained using both boundary conditions. We have noted that in order to minimize the influence on the streamer propagation it is necessary to have a much larger computational domain using the homogeneous Neumann boundary condition ($\vec{\nabla} V \cdot \vec{n} = 0$)

than using the Dirichlet boundary condition based on Laplace's equation. Then in this work, we have used a Dirichlet boundary condition based on the solution of Laplace's equation.

2.2. Analytical solution of Laplace's equation in point-to-point and point-to-plane geometries

As mentioned in the previous section, the analytical solution of Laplace's equation can be used to define boundary conditions. It is also interesting to use this analytical solution to initialize the electric potential in the whole computational domain to derive the Laplacian potential at $t = 0$ if an iterative solver for Poisson's equation is used.

For the point-to-plane and point-to-point geometries, we consider that point electrodes are infinite hyperboloids and that in the point-to-point geometry, electrodes are branches of the same hyperboloid. As shown in figure 3 in this case, the geometrical configuration has a planar symmetry. In experiments, the length of electrodes is obviously finite. Then, if the absolute values of the electric potential applied to the anode and to the cathode are different (e.g. a grounded cathode and the anode set to $V = V_a$), the electric field lines starting from the anode do not necessarily converge at the cathode and can go to infinity. Thus, both electrodes are not in 'total influence', and then the system is no longer symmetric. In this case, the electric field in the vicinity of the anode is higher than the one in the vicinity of the cathode. To obtain equal electric fields, one can roughly estimate that the gap length between the two electrodes has to be less than or equal to a characteristic length of the electrodes (e.g. radius of a sphere, length of a hyperboloid, etc).

From an electrostatic point of view, the point-to-plane configuration is very close to the point-to-point configuration. Indeed, since the geometrical configuration of the two electrodes is symmetric, the plane of symmetry in the middle of the gap is an equipotential of the electrostatic system (see figure 3). Then one can adjust this equipotential to any value by changing the potential of one of the electrodes.

Given the geometry of the system, it is useful to work in hyperbolic coordinates:

$$\begin{aligned} x &= \alpha \xi (1 + \eta^2)^{1/2}, \\ r &= \alpha \eta (1 - \xi^2)^{1/2}, \end{aligned} \quad (13)$$

where $\xi^2 < 1$ and $\alpha \neq 0$. As in figure 3 the electrodes are two branches of the same hyperboloid; the surfaces of both needle electrodes are defined by the Cartesian equation:

$$\frac{x^2}{a^2} - \frac{r^2}{b^2} = 1. \quad (14)$$

Using equation (13), we have $a^2 = \alpha^2 \xi^2$, $b^2 = \alpha^2 (1 - \xi^2)$ and the squared focal length of the hyperbola is $a^2 + b^2 = \alpha^2$. In such a system of coordinates, a hyperbola with the focal length α is defined by the equation $\xi = C$, where C is a constant, and the electrode surface is defined by $\xi_0^2 = a^2/\alpha^2$. For this geometry, the electric potential has the following form (Eyring *et al* 1928):

$$V(\xi) = \frac{A}{2} \log \frac{1 + \xi}{1 - \xi} + B, \quad (15)$$

where A and B are constants depending on the electric potentials of the electrodes and their shapes. It is interesting to note that V depends only on ξ , which means that equipotential surfaces of the system are branches of confocal hyperbolas. In point-to-point geometry, one can take the symmetric electrode surfaces $\xi = \xi_0$ for the anode of potential V_a and $\xi = -\xi_0$ for the cathode at potential V_c ; then one obtains

$$\begin{aligned} A &= \frac{V_a - V_c}{\log\left(\frac{1 + \xi_0}{1 - \xi_0}\right)}, \\ B &= \frac{V_a + V_c}{2}. \end{aligned} \quad (16)$$

To find A and B in the point-to-plane system, where the point is a hyperboloid and the plane is a grounded cathode, it is just necessary to set $V_c = -V_a$ in equation (16). Then $V = 0$ represents an equipotential plane corresponding to the cathode plane defined by $\xi = 0$. Finally, to easily use equation (15) on a rectilinear grid, we have derived an expression for ξ in cylindrical coordinates as

$$\xi^2(x, r) = \frac{\alpha^2 + x^2 + r^2 - [(\alpha^2 + x^2 + r^2)^2 - 4\alpha^2 x^2]^{1/2}}{2\alpha^2}. \quad (17)$$

Note that from equation (13), $\xi < 0$ (respectively, $\xi > 0$) corresponds to $x < 0$ (respectively, $x > 0$).

For streamer applications, it is also interesting to derive the value of the magnitude of the electric field. Using the hyperbolic coordinates given by equation (13), we have (Eyring *et al* 1928)

$$\vec{\nabla} V = \frac{1}{\Xi} \frac{\partial V}{\partial \xi} \vec{u}_\xi, \quad (18)$$

where

$$\vec{u}_\xi = \frac{\vec{\nabla} \xi}{\|\vec{\nabla} \xi\|}$$

and

$$\Xi(\eta, \xi) = \alpha \left(\frac{\eta^2 - \xi^2 + 1}{1 - \xi^2} \right)^{1/2}. \quad (19)$$

Using equations (13) and (17), one can easily derive $\Xi(x, r)$. Finally, the magnitude of the electric field can be derived from equation (15) as

$$|\vec{E}(x, r)| = \frac{1}{\Xi(x, r)} \frac{A}{1 - \xi^2(x, r)}. \quad (20)$$

2.3. Numerical methods

In order to compare our results in point-to-plane geometry with those obtained by Kulikovsky (1998), we have used the same numerical scheme (i.e. the modified Scharfetter–Gummel algorithm (Kulikovsky 1995)) to solve the charged species transport equations. This numerical scheme has also been used for streamer simulations in point-to-point geometries in section 3.3. For both geometries, close to the needle electrodes, no special numerical treatment has been implemented to take into account the exact shape of the electrodes in setting

up the boundary conditions for drift–diffusion equations of charged species. The boundary conditions are computed in both directions simply assuming that the interface has a ‘staircase’ shape. In this work, as in Kulikovsky (1997a) we have taken into account simplified boundary conditions: near the anode and cathode surfaces, gradients of electron density are assumed to be zero. In Kulikovsky (1998), the finite difference form of Poisson’s equation is solved using the symmetrical successive overrelaxation (SOR) method. This method is simple and robust and has been widely used for streamer simulations. However, the SOR method is well known to converge rather slowly. To reduce computation times, in Bourdon *et al* (2007), we have used the D03EBF module of the NAG Fortran library (<http://www.nag.co.uk>) which is based on the iterative Stone algorithm (Stone 1968) and converges more rapidly than the SOR method for the same accuracy. It is interesting to note that recently in the literature different efficient direct solvers (e.g. MUMPS (Amestoy *et al* 2000, 2001, 2006) and SuperLU solver (Demmel *et al* 1999a, 1999b) (<http://crd.lbl.gov/~xiaoye/SuperLU/>)) have been developed for large systems of linear equations and can be applied to solve Poisson’s equation. Then, as part of the preparatory work for the studies presented in this paper, we have conducted several test runs to compare the results obtained using the iterative NAG module and the direct SuperLU and MUMPS solvers (Bonaventura *et al* 2008). It turned out that there are only minor changes in the streamer dynamics computed using these two solvers. Direct solvers are inherently very accurate and robust, and furthermore for the case of a fixed grid, the most time-consuming step in the calculation of the solution (i.e. the factorization) needs to be done only once at the beginning of the discharge simulation. Therefore, at each timestep of the discharge simulation, the calculation of the solution for different source terms (i.e. term on the right-hand side of equation (6)) is very fast. The SuperLU solver was adopted to solve Poisson’s equation for all simulation studies presented in this paper.

We have followed the approach discussed in Vitello *et al* (1994) to select the timestep for our streamer simulation. This approach is based on the calculation of three time scales: the Courant δt_c , effective ionization δt_i and dielectric relaxation δt_D time scales. The model timestep is calculated as $\delta t = \min(A_c \delta t_c, A_i \delta t_i, A_D \delta t_D)$ with $A_c = 0.5$, $A_i = 0.05$ and $A_D = 0.5$. We note that in this work, as in Bourdon *et al* (2007), we adopt a value of A_i which is a factor of 2 less than that used in Vitello *et al* (1994).

To calculate the photoionization source term in the streamer model, we use the three-group SP₃ model presented in Bourdon *et al* (2007) and successfully applied to the simulation of the propagation of a double-headed streamer in strong external electric fields (i.e. higher than the conventional breakdown threshold field defined by the equality of the electron impact ionization and electron dissociative attachment coefficients in air) and in Liu *et al* (2007, 2008) to the simulation of a streamer propagation in a weak external electric field. We use boundary conditions derived from the radiative transfer equation (Liu *et al* 2007).

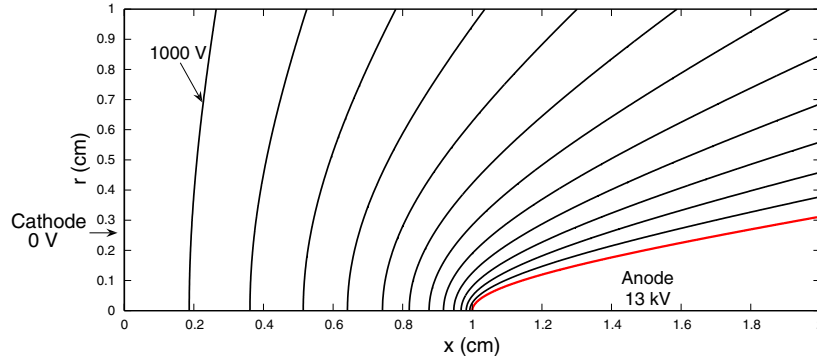


Figure 4. Isocontours of the electric potential in a point-to-plane configuration. The isocontours are shown from 0 V (cathode) to 13 000 V (anode), with a step of 1000 V. In this figure, isocontours of the analytical solution and of the numerical simulation with the GFM overlap, demonstrating good agreement between the two solutions.

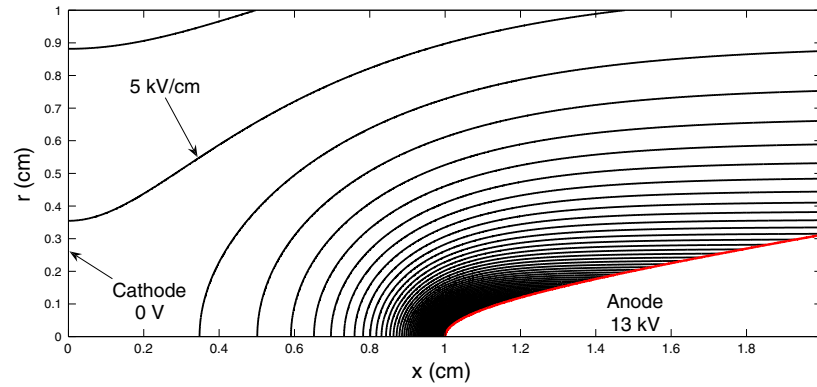


Figure 5. Isocontours of the magnitude of the electric field for the same conditions as figure 4. The isocontours are shown from 1 to 160 kV cm⁻¹, with a step of 1 kV cm⁻¹. In this figure, isocontours of the analytical solution and of the electric field computed using the GFM overlap, demonstrating good agreement between the two solutions.

3. Results and discussion

3.1. Laplacian field in a point-to-plane geometry

In this section, a point-to-plane configuration is used to compare the electric potential calculated in the whole computational domain using the GFM presented in section 2.1 and the analytical solution presented in section 2.2. We have considered the same configuration as in Kulikovsky (1998): a hyperboloid anode ($V_a = 13$ kV) is located at $x = 1$ cm and a plane grounded cathode (i.e. $V = 0$) is located at $x = 0$. Parameters of the shape of the hyperboloid anode are the same as in Kulikovsky (1998): $a = 1$ cm and $b = 0.18$ cm which correspond to a radius of curvature at the tip of the electrode of $b^2/a = 324 \mu\text{m}$. The computational domain dimension is 2×1 cm² and is discretized on a Cartesian grid in both radial and axial directions. The total number of cells is $n_x \times n_r = 1000 \times 500$, where n_x and n_r represent the number of cells in the axial and radial directions, respectively. This corresponds to cell sizes of $20 \mu\text{m}$ in both the axial and radial directions. For streamer simulations, the typical cell size is much smaller in the region of the streamer head and is on the order of $5 \mu\text{m}$ or less. In this section, we propose to use a large grid size to clearly demonstrate the efficiency of the GFM.

Figure 4 compares the isocontours of the electric potential calculated using the GFM and the analytical solution given

by equation (15). Very good agreement between both results is observed at boundaries and in the whole computational domain. Small variations of the electric potential can lead to significant changes in the electric field. Figure 5 compares the analytical solution of the electric field (equation (20)) with the electric field computed with the GFM, for the same condition as figure 4. In figure 5, very good agreement is observed between both results at boundaries and in the computational domain. However this figure does not show the good agreement of results close to the anode tip where the contour lines overlap due to the rapid variation of the electric field. In order to show the quantitative improvement due to the use of the GFM close to the electrode, we compare in figures 6 and 7 the magnitude of the electric field computed, on the same grid, with the GFM and without any specific treatment to take into account the real shape of the electrode (i.e. ‘the staircase approach’).

On the symmetry axis, figure 6 shows that the electric field calculated using the GFM is in very good agreement with the analytical solution close to the electrode. The results obtained with the staircase approach on the same grid overestimate the electric field in the vicinity of the electrode. We have checked that the discrepancy between the two numerical approaches increases as the grid becomes coarser.

Figure 7 shows in the whole computational domain the magnitude of the electric field obtained with and without the GFM. We note that the results obtained without the GFM

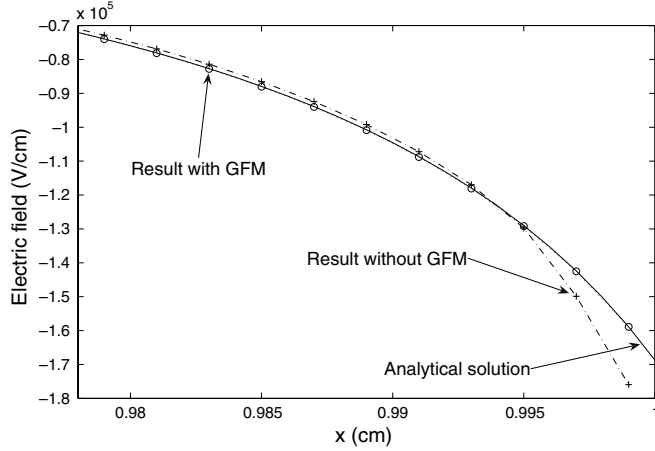


Figure 6. Norm of the electric field along the symmetry axis in the vicinity of the anode tip. Solid line: analytical solution (equation (20)). Dotted line with circles: numerical results using the GFM. Dotted-dashed line with crosses: numerical results without the GFM (i.e. ‘staircase’ approach). The symbols for the numerical results correspond to the grid nodes.

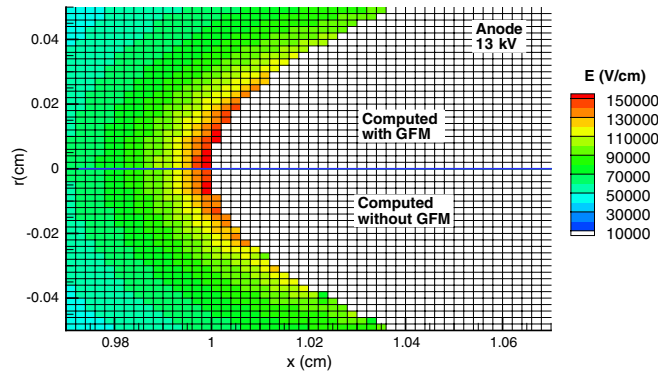


Figure 7. Comparison between the magnitude of the electric field computed with and without the GFM in the same point-to-plane configuration as figure 4. The white region of the colour scale corresponds to the hyperboloid anode. In the case without the GFM all the potentials $V_{i,j}$ are set equal to V_a if the point (i, j) is inside the electrode. In the case with the GFM the nodes inside the electrode are also set to V_a , except for the points close to the interface.

present oscillations of the electric field along the electrode interface due to the random crossing of the electrode over the cells. Furthermore, it is important to mention that with this grid the tip of the electrode appears to be flat and then the maximum of the electric field is not located on the symmetry axis but a few cells below (see figure 7). We have carried out simulations of streamer propagation with both Laplacian electric fields and we have observed that the oscillations in the electric field calculated without using the GFM may lead to streamer branching close to the electrode. Using the GFM on the same grid, no streamer branching has been observed.

3.2. Positive streamer propagation in a point-to-plane geometry

In this section, we apply the GFM to the simulation of a positive streamer propagation in the point-to-plane geometry studied by Kulikovsky (1998). The parameters of the shape

of the hyperboloid anode, the location of the electrodes and the applied voltage are the same as in section 3.1. We have used the same transport parameters and reaction rates as in Kulikovsky (1998). For the photoionization source term in air, we have also used the model proposed by Zheleznyak *et al* (1982). However, in Kulikovsky (1998), this source term is calculated using an integral approach, which is very time consuming. In our work, we have implemented the photoionization source term using the efficient differential three-group SP_3 approach (Bourdon *et al* 2007).

If the integral model is used without any geometric limitations on the emitting volume of the discharge, we have shown in Bourdon *et al* (2007) that the results of both approaches are in very good agreement, but the computation time of the three-group SP_3 model can be up to three orders of magnitude smaller. In Kulikovsky (1998), the computational domain dimension is $6 \times 1.1 \text{ cm}^2$ and is discretized on a grid with $n_x \times n_r = 640 \times 134$ points. In the axial direction, 200 uniform cells form the coarse grid. To resolve the steep gradients accurately, the streamer head is covered by a moving window with a uniform fine grid (of $\simeq 400$ cells) with cell sizes of $6.3 \mu\text{m} \times 8 \mu\text{m}$. In this work, we have used a fixed grid. Different tests have been carried out on the grid to reduce the size of the computational domain and to limit the number of grid points. Finally, in this work we have used a $2 \times 0.5 \text{ cm}^2$ computational domain discretized on a fixed rectilinear grid with $n_x \times n_r = 2254 \times 344$ cells. The grid is Cartesian, with a fixed cell size of $5 \mu\text{m}$ in the region where the streamer propagates. Beyond this region (i.e. $x \in [1.1, 2] \text{ cm}$ and $r \in [0.15, 0.5] \text{ cm}$) the grid expands following a geometric progression. As in Kulikovsky (1998), a neutral plasma spot is placed at the anode tip to initiate streamer formation:

$$n_p(x, r)|_{t=0} = n_e(x, r)|_{t=0} = n_0 \exp \left[-\frac{r^2}{\sigma_r^2} - \frac{(x - x_0)^2}{\sigma_x^2} \right], \quad (21)$$

where $x_0 = a - \sigma_x$, $\sigma_x = 0.025 \text{ cm}$ and $\sigma_r = 0.01 \text{ cm}$. At $t = 0$ there are no negative ions. The initial maximum density is $n_0 = 10^{14} \text{ cm}^{-3}$.

Figures 8 and 9 show the same results as figures 2 and 3 in Kulikovsky (1998). Figure 8 shows the contour lines of the electron density and absolute values of the electric field during the positive streamer propagation towards the cathode. Figure 9 shows the evolution of axial profiles of the electron density and electric field. For both figures, we note a very good agreement with the results from Kulikovsky (1998). Only slight differences are observed in the contour lines. In figure 8, we note that in our results the contour line for 30 kV cm^{-1} is slightly larger in the radial direction and slightly shorter in the axial direction than in the results of Kulikovsky (1998). Direct inspection of the contour lines of the electron density indicates that the variation of the thickness of the streamer channel is smoother in our results than in the results from Kulikovsky (1998). At $t = 1 \text{ ns}$, we note that the field in the streamer head is higher in our results than in the results of Kulikovsky (1998). It is interesting to note that, as part of the preparatory work for the studies presented in this paper, we have also carried out simulations using the staircase approach

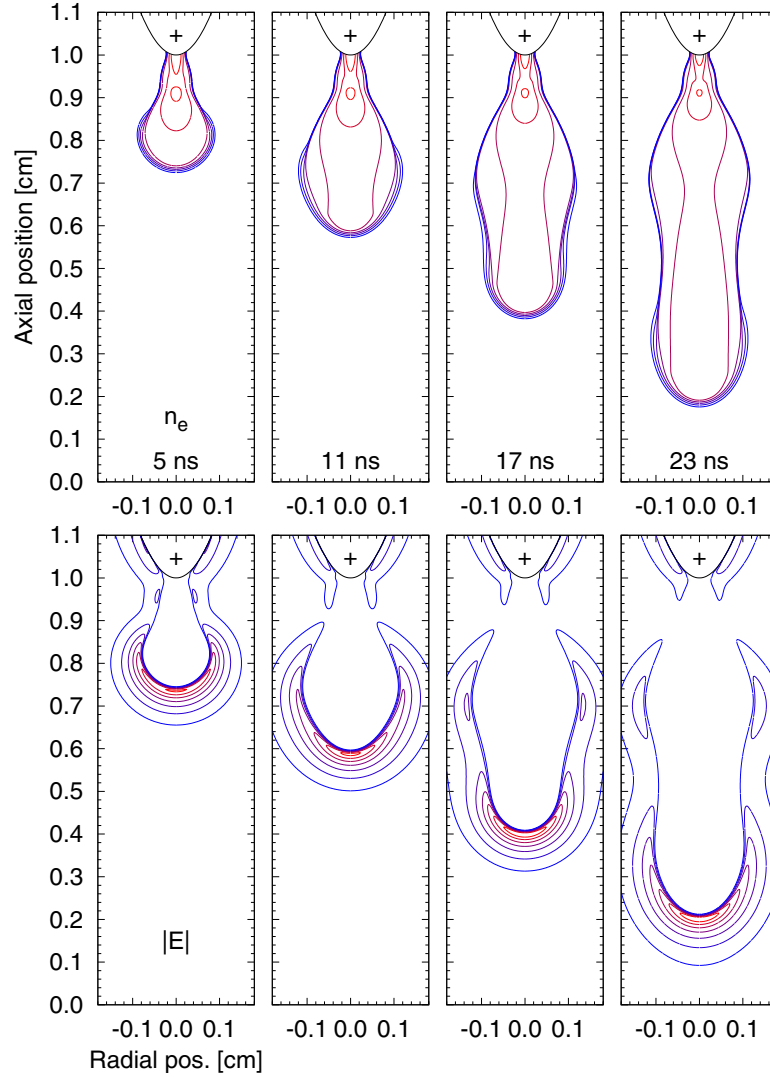


Figure 8. Two-dimensional isocontours of the electron density and the magnitude of the electric field for the moments of time 5, 11, 17 and 23 ns. For the electron density the isocontours go from 10^{11} to 10^{14} cm^{-3} , with a multiplier step of $10^{0.5} \text{ cm}^{-3}$. For the electric field the isocontours go from 30 to 90 kV cm^{-1} , with a step of 10 kV cm^{-1} .

on the fixed grid used for the GFM simulations, and we obtained at $t = 1 \text{ ns}$ a field close to the one obtained by Kulikovsky (1998). Then this field difference may be due to the fact that the GFM is not used in Kulikovsky (1998). In figures 8 and 9, the two phases of the streamer propagation can be clearly observed. The first one lasts about 5 ns and corresponds to the positive streamer formation close to the point anode and to its propagation in the high field region close to the electrode tip. We note that during this first phase, the streamer radius increases as the streamer propagates towards the cathode. For $t > 5 \text{ ns}$, the streamer propagates with almost constant velocity, radius of the plasma channel and maximum electric field in the streamer head. In Kulikovsky (1998), this phase is called the stationary propagation phase of the streamer. The very good agreement observed between our results and those obtained by Kulikovsky (1998) seems to indicate that the use of a staircase approach or the GFM around the point electrode has only a weak influence on the streamer results for this test case, as long as the grid is sufficiently refined close to the point electrode when the

streamer starts to propagate. These results are nevertheless of great interest since they clearly show that the GFM can be used for streamer simulations. It is important to note that the GFM is easy to implement and guarantees an accurate calculation of the electric field close to the electrodes, conversely to a staircase approach. Thus, this method enables the use of coarser grids to define the electrode. The GFM also avoids the numerous parametric studies on numerical grids required to use the staircase approach without notable numerical instabilities occurring.

3.3. Streamer propagation in point-to-point geometry

In this section, we apply the GFM to simulate a discharge in air at atmospheric pressure between two point electrodes. This discharge has been studied experimentally in Pai *et al* (2008), Pai (2008). These authors use a nanosecond repetitively pulsed (NRP) voltage (5–10 kV) to generate a discharge in a preheated air flow between two pin electrodes separated by 5 mm. In the experimental study, the exact shape of the electrodes is not

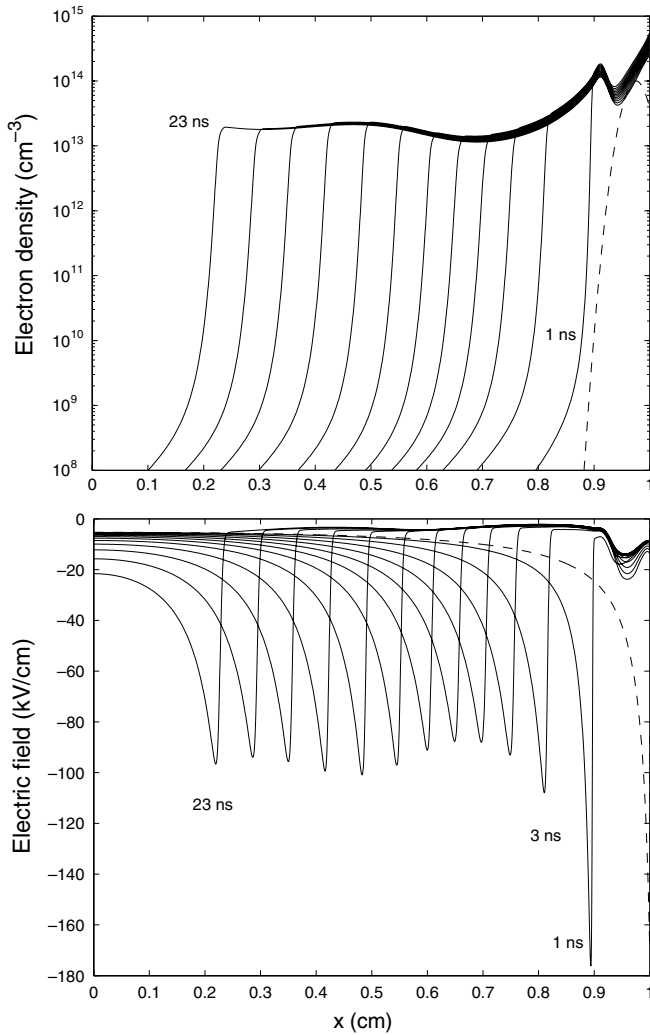


Figure 9. Electron density and electric field on the axis of symmetry at time moments from 1 to 23 ns with a timestep of 2 ns. Dashed lines: electron density and electric field at $t = 0$.

known very accurately. Nevertheless, the radius of curvature of both needle electrodes is about $200\ \mu\text{m}$ and then, as a first approximation, we have considered that both electrodes are hyperboloids with a radius of curvature of $324\ \mu\text{m}$ as in sections 3.1 and 3.2. It is interesting to note that the cumulative effect of repeated pulsing achieves steady-state behaviour and then, even if they are transient, the NRP discharges have a visual resemblance to dc discharge regimes. For an air temperature of 1000 K, as the applied voltage increases, three different discharge regimes are observed. At a low voltage, the ‘C’ regime, like a corona discharge, emits light only near the anode. For voltages higher than 6 kV, the ‘D’ regime is observed. This regime has an emission which fills the gap in a diffuse manner. Finally, for voltages higher than 7 kV, an ‘F’ regime is observed. This regime is reminiscent of a spark discharge with an intense emission. The measurements show that the ‘D’ regime has low levels of emission, gas heating and electrical conduction current, indicating that it is non-thermal. Furthermore this regime develops through a positive streamer propagating between both electrodes followed by a return wave in the opposite direction. In contrast to the non-thermal ‘D’

regime, the thermal ‘F’ regime emits strongly, heats the gas several thousand degrees Kelvin and has tens of amperes of conduction current. Further details on the experimental results may be found in [Pai \(2008\)](#).

The simulation of successive discharges due to a high frequency repetitively nanosecond pulsed voltage is beyond the scope of this paper. In this section, we propose to apply the GFM to simulate a single discharge in air occurring during one of the 10 ns long voltage pulses. We have considered an applied voltage of 7 kV which corresponds to the highest value of voltage for which the ‘D’ regime is observed. The velocity of the air flow is $1\ \text{m s}^{-1}$ and then, we have assumed that air is quiescent during one discharge pulse and is uniformly preheated at 1000 K. As many discharges have occurred before the one we simulated in this work, it is necessary to estimate the seed charges in the interelectrode gap. For a 10 kHz voltage frequency, the time between two voltage pulses is $10^{-4}\ \text{s}$. As air is preheated at 1000 K, the attachment processes can be neglected and then the concentration of seed charges is mainly controlled by recombination processes. For the conditions we want to simulate, the recombination time has been estimated to be on the order of $10^{-4}\ \text{s}$. Then, based on the work of [Pancheshnyi \(2005\)](#), we have assumed that the density of seed charges before each new pulse is in the 10^9 – $10^{10}\ \text{cm}^{-3}$ range. For the calculations presented in this section a value of $10^9\ \text{cm}^{-3}$ has been used. As a first approximation, we have assumed that these seed charges are distributed uniformly in the computational domain. Due to this relatively high pre-ionization level, we have checked that photoionization has no influence on the results presented in this section. To take into account the fact that the discharge occurs at atmospheric pressure in preheated air at $T = 1000\ \text{K}$, we have simply changed the value of the total density $N = N_0 T_0 / T$, where $N_0 = 2.45 \times 10^{25}\ \text{m}^{-3}$ is the air neutral density at ground pressure and ambient temperature ($T_0 = 300\ \text{K}$). This decrease in the total density by a factor of 3 increases by the same factor the local reduced electric field E/N and then has a direct impact on transport parameters and reaction rates in air which are assumed to be functions of E/N .

In this work, we have used a $2 \times 0.5\ \text{cm}^2$ computational domain discretized on a fixed rectilinear grid with $n_x \times n_r = 1510 \times 347$ cells. The grid is Cartesian, with a fixed cell size of $5\ \mu\text{m}$ in the region where the streamer propagates. Beyond this region the grid expands following a geometric progression. The tip of the anode ($V_a = 7\ \text{kV}$) is located at $x = 1.25\ \text{cm}$ and the tip of the grounded cathode is located at $x = 0.75\ \text{cm}$.

Figure 10 shows the evolutions of the contour lines of the electron density and absolute values of the electric field. Figures 11 and 12 show the evolutions of axial profiles of the electric field and the electron density, respectively. Two phases of the discharge evolution can be clearly distinguished. The first one lasts about 5 ns. During this time period, two discharges are initiated at both electrode tips where the Laplacian electric field is highest (i.e. $70\ \text{kV cm}^{-1}$) and these discharges propagate in the gap. On the anode side, the conditions for positive streamer propagation are fulfilled in the first nanosecond and a positive streamer propagates towards the grounded cathode. The electron density in the streamer

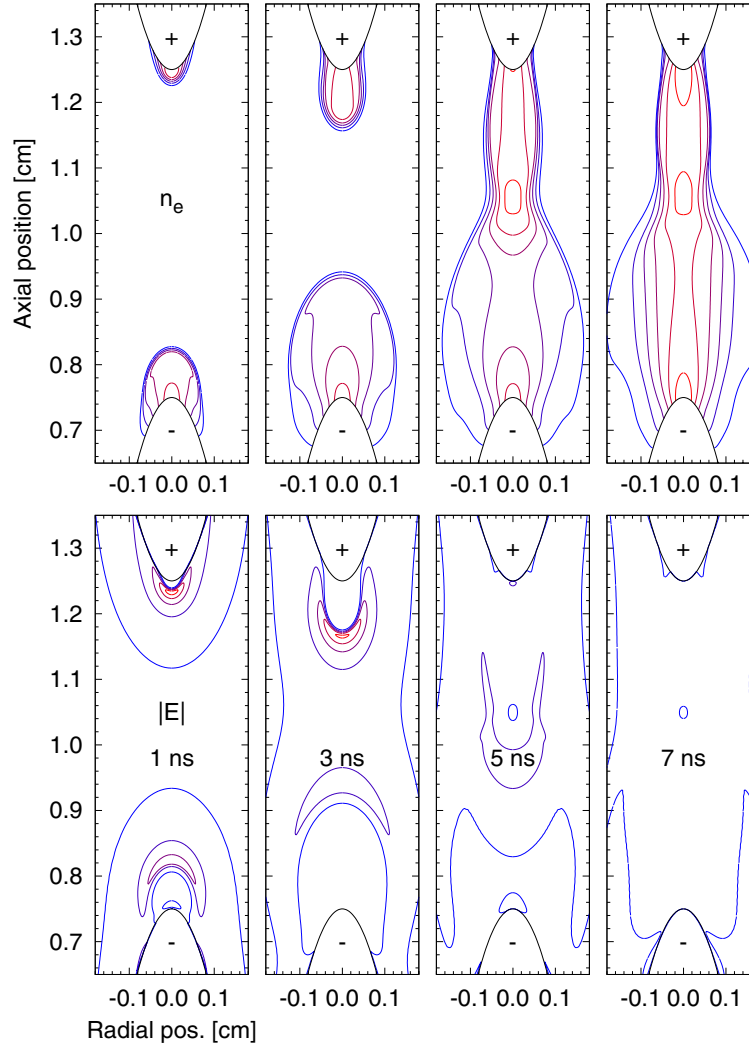


Figure 10. Point-to-point geometry: two-dimensional isocontours of the electron density and the magnitude of the electric field for the moments of time 1, 3, 5 and 7 ns. For the electron density the isocontours go from 10^{11} to $10^{13.5} \text{ cm}^{-3}$, with a multiplier step of $10^{0.5} \text{ cm}^{-3}$. For the electric field the isocontours go from 10 to 50 kV cm^{-1} , with a step of 10 kV cm^{-1} .

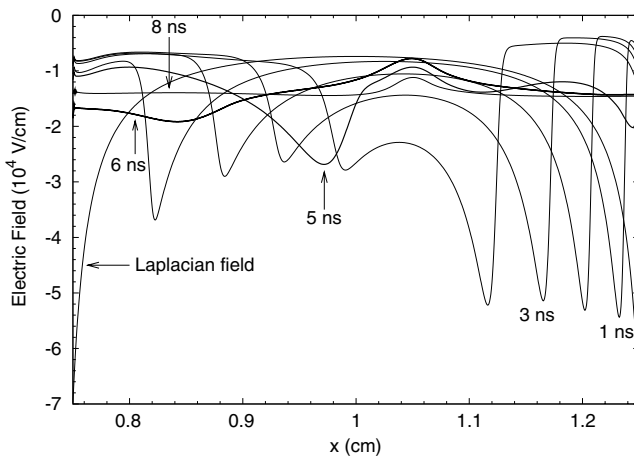


Figure 11. Point-to-point geometry: electric field on the axis of symmetry for time moments from 0 to 8 ns.

channel is about $1.5 \times 10^{13} \text{ cm}^{-3}$ and the maximum electric field on the streamer axis is 50 kV cm^{-1} . It is interesting to note that these values are less than those obtained for the positive streamer simulated in section 3.2. In fact, due

to pre-ionization in the gap and the lower total density, the conditions for positive streamer propagation are much more favourable than those of section 3.2 and then the streamer propagates with a lower electric field and a lower electron density in its head. These results are in agreement with those obtained by Liu and Pasko (2006). On the cathode side, in the first few nanoseconds we observe the expansion of a discharge towards the anode. The maximum electric field in the discharge decreases as the discharge propagates towards the anode and is on the order of 30 kV cm^{-1} and the electron density in the discharge front is on the order of 10^{12} cm^{-3} . Due to pre-ionization in the interelectrode gap these values are less than those usually obtained for negative streamers (e.g. Bourdon *et al* 2007), and we have considered in this work that the discharge which propagates from the cathode is a ‘weak’ negative streamer. It is interesting to note in figure 10 that both discharges have very different radial expansions. At $t \simeq 5 \text{ ns}$, both discharges interact at $z \simeq 1.05 \text{ cm}$ and figures 11 and 12 clearly show that the positive streamer propagates very rapidly towards the cathode in the volume pre-ionized by the ‘weak’ negative streamer. Then at $t = 7 \text{ ns}$ the positive streamer reaches the grounded electrode and figure 11 shows

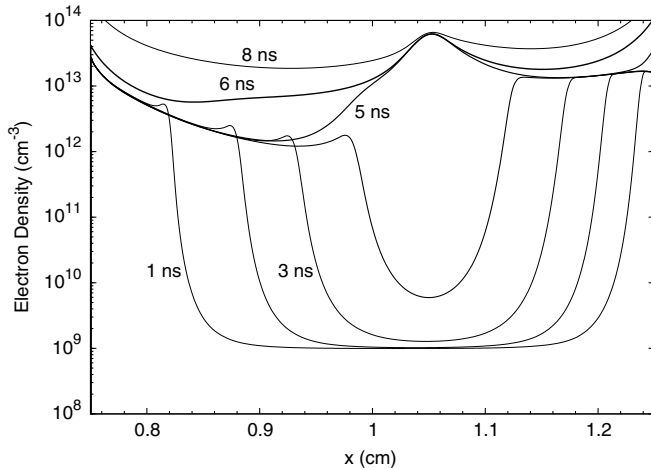


Figure 12. Point-to-point geometry: electron density on the axis of symmetry for time moments from 1 to 8 ns.

the rapid redistribution of the potential in the interelectrode gap. In figure 12, we note that the interaction between the two discharges creates a bump in the electron density at $z \simeq 1.05$ cm and this bump remains nearly unchanged for $t > 5$ ns. We also note that for $t > 5$ ns the electron number density increases on the discharge axis.

All these results put forward the fact that a positive streamer could propagate between the two point electrodes in less than 7 ns, which is in qualitative agreement with the experimental results. The results of a parametric study on the influence of the voltage, the radius of curvature of the electrodes, the interelectrode gap and the gas temperature on the structure of the discharge in point-to-point geometry will be presented in a separate dedicated paper with a detailed comparison with experiment.

4. Conclusions

In this paper, we present the application of the GFM to the simulation of streamer discharges between electrodes of complex geometries. In streamer simulations, it is particularly important to calculate the electric field accurately since, first, it is directly related to charged species densities in Poisson's equation and, second, transport parameters and source terms have a strong non-linear dependence with the electric field. Then in this work, we have used the GFM to solve Poisson's equation to calculate accurately the potential and electric field close to the electrode. This method allows one to take into account the influence of the exact shape of the electrodes in a rectilinear grid.

Three numerical solvers for Poisson's equation have been tested in this work: the iterative NAG module and the direct SuperLU and MUMPS solvers. Only minor discrepancies have been observed in the streamer dynamics computed using these two solvers. In fact, direct solvers are inherently very accurate and robust and furthermore for the case of a fixed grid, the most time consuming step of the calculation of the solution (i.e. the factorization) needs to be done only once at the beginning of the discharge simulation. The direct SuperLU solver was

adopted to solve Poisson's equation for all studies presented in this paper.

To validate and test the performances of the GFM to solve Poisson's equation for streamer simulations, first, we have compared in a point-to-plane geometry, the analytical solution for the potential and Laplacian electric field with the results calculated with the GFM. Very good agreement has been observed. To clearly show the improvement due to the use of the GFM close to the electrode, we have also compared, on the same grid, the magnitude of the electric field computed with the GFM and without any specific treatment to take into account the real shape of the electrode (i.e. 'the staircase approach'). As expected, the discrepancy between the two numerical approaches increases as the grid becomes coarser. We have also shown that without the GFM, the electric field along the electrode interface may present unphysical oscillations due to the random crossing of the electrode surface with the grid. Second, we have compared results obtained using the GFM with the results obtained by Kulikovsky (1998) for the simulation of the positive streamer propagation in a hyperboloid-to-plane configuration. Very good agreement has been obtained on all positive streamer characteristics during its propagation in the interelectrode gap. Finally, we have applied the GFM to simulate the discharge in preheated air (at 1000 K) at atmospheric pressure in point-to-point geometry (with a 5 mm interelectrode gap). This discharge has been studied experimentally in Pai *et al* (2008), Pai (2008). For a constant applied voltage of 7 kV, we have shown that positive and negative streamers start to propagate in the gap starting from the anode and the cathode, respectively. After the interaction of both discharges in the gap, we have observed a very rapid propagation of the positive streamer towards the cathode in the volume pre-ionized by the negative streamer. This structure and the propagation time scale of the discharge are in qualitative agreement with the experiment. However, this study is a first step towards a more thorough comparison study between experiment and simulation. In particular, based on simulations, it would be important to better understand the influence of the voltage, the radius of curvature of the electrodes, the interelectrode gap and the gas temperature on the structure of the discharge in point-to-point geometry. Furthermore, in this work, we have studied the streamer propagation dynamics assuming a constant gas temperature and the Joule heating of the gas by the discharge has not been taken into account. In a further work, it would be interesting to take into account the gas heating to better understand the transition between the non-thermal 'D' and the thermal 'F' regimes.

In this work, we have shown that the GFM is a simple method, easy to implement in existing discharge codes. In this paper, we have studied discharges propagating between metallic electrodes but the GFM can be easily extended to simulate dielectric barrier discharges in complex two- and three-dimensional geometries or to study the influence of obstacles (i.e. dust particles, droplets, etc) on the streamer path.

In this work, no special numerical treatment has been done close to the needle electrodes to take into account the exact shape of the electrode in setting up boundary conditions for

drift–diffusion equations of charged species. The boundary conditions are computed in both directions assuming that the interface has a ‘staircase’ shape. In this paper, as in Kulikovsky (1997a), we have taken into account simplified boundary conditions: near the anode and the cathode surface, gradients of electron density are taken equal to zero. To take into account more realistic boundary conditions on the electrodes of complex geometries (i.e. secondary emission processes due to ion bombardment, photoemission, etc), it is necessary to model more accurately the real shape of the electrode for the writing of boundary conditions of drift–diffusion equations of charged species. For flows in complex geometries, Tseng and Ferziger (2003) have developed the ghost-cell immersed boundary method (GCIBM). This method allows one to treat Dirichlet and Neumann boundary conditions while preserving the overall second-order accuracy of the base solver they use in the flow far from boundaries. In Zeghondy *et al* (2007), we have applied the GCIBM to implement Neumann boundary conditions on a hyperboloid anode in a point-to-plane geometry. The simultaneous use of the GFM for Poisson’s equation and the GCIBM for drift–diffusion equations to simulate discharge in point-to-plane and point-to-point geometries with different types of boundary conditions for drift–diffusion equations will be presented in a separate dedicated paper.

Acknowledgments

This work was supported by the Agence Nationale de la Recherche (ANR) under the ‘Interaction Plasma Ecoulement Réactif (IPER)’ project and by Ecole Centrale Paris. Z Bonaventura is also grateful to the Ministry of Education of the Czech Republic under contract MSM0021622411. The authors would like to thank Drs David Pai, Deanna Lacoste and Professor Christophe Laux (EM2C, Ecole Centrale Paris) for helpful discussions on the air discharge structure in point-to-point geometry.

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